

Statistics Test Review: Normal Distributions, Samples & Data Collection

Course: Advanced Algebra (Larson & Boswell): Data Analysis and Statistics.

How to use this: Part 1 is the one-page formula and vocabulary sheet. Parts 2–8 follow the test topics in order, each with short notes, a comparison table where it helps, and fully solved examples. The standard normal table is at the very end.

Calculator: answers match both the table and a TI-84 (`normalcdf` / `invNorm`). The table is always enough.

Part 1 — Formula & Vocabulary Sheet

Formulas

Name	Formula	What it means
Mean of a population	$\mu = (\Sigma x) / N$	Add all values, divide by how many there are. μ is read “mu”.
Mean of a sample	$\bar{x} = (\Sigma x) / n$	Same calculation, but for a sample. $\bar{x}$ is read “x-bar”.
Standard deviation	$\sigma = \sqrt{[\Sigma (x - \bar{x})^2 / n]}$	Typical distance of the data from the mean. Small σ means the data is bunched together; large σ means it is spread out.
z-score	$z = (x - \mu) / \sigma$	How many standard deviations a value x is above (+) or below (–) the mean.
Value from a z-score	$x = \mu + z \cdot \sigma$	Undo a z-score to get back to the real value.
Empirical rule	68% within $\mu \pm \sigma$, 95% within $\mu \pm 2\sigma$, 99.7% within $\mu \pm 3\sigma$	Only for normal (bell-shaped) data.
Area to the right	$P(z \geq a) = 1 - (\text{table value at } a)$	The table gives area to the left, so subtract from 1.
Area between	$P(a \leq z \leq b) = (\text{table at } b) - (\text{table at } a)$	Big left area minus small left area.
Margin of error (sample proportion)	$\pm 1 / \sqrt{n}$	For a random sample of size n , the true population percentage is likely within this much of the sample percentage.
Interval for the true value	sample % – $1/\sqrt{n}$ to sample % + $1/\sqrt{n}$	Where the population value probably lies.

Quick worked check of the standard deviation. Quiz scores: 6, 7, 7, 8, 8, 8, 8, 9, 9, 10.

- Mean: $80 \div 10 = 8$.
- Squared distances from 8: 4, 1, 1, 0, 0, 0, 0, 1, 1, 4, which add to 12.
- $\sigma = \sqrt{(12 \div 10)} = \sqrt{1.2} \approx 1.10$.

On a TI-84 (STAT → CALC → 1-Var Stats) this shows as $\sigma x = 1.10$. The $Sx = 1.15$ line divides by $n - 1$ instead; use σx unless your teacher says otherwise.

Vocabulary

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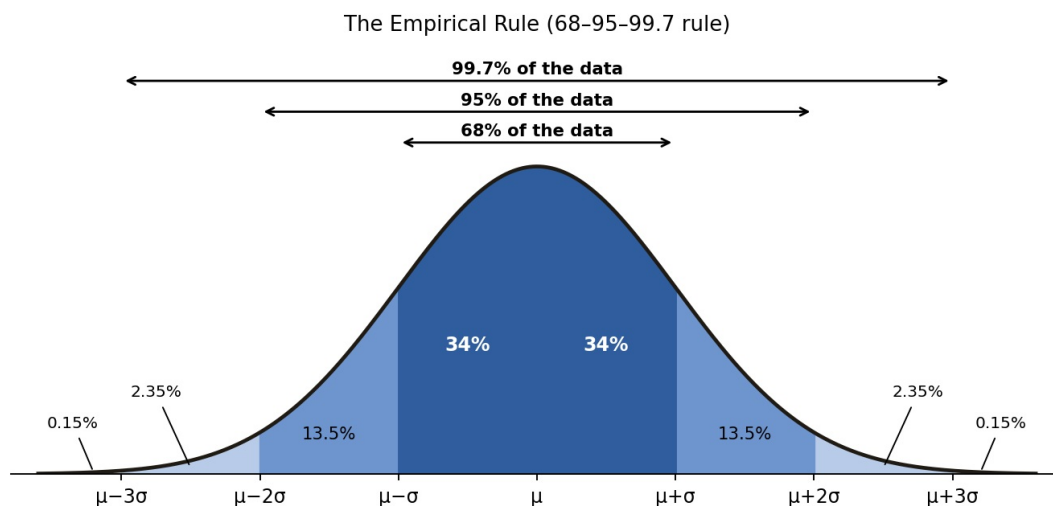
Term	Definition
Normal distribution	A bell-shaped, symmetric distribution. Its mean, median and mode are all at the center, and its spread is described by σ .
Normal curve	The smooth bell-shaped graph of a normal distribution. The total area under it is 1 (100%).
Standard normal distribution	The normal distribution with mean 0 and standard deviation 1. Any normal data turns into it when you convert to z-scores.
Standard normal table	A table that gives the area under the standard normal curve to the left of a z-score, which is the probability of getting a value at or below it.
Percentile	The percent of data at or below a value. A score at the 90th percentile is higher than 90% of scores.
Skewed	Not symmetric: one tail is longer. Skewed right means the long tail is on the right (mean > median). Skewed left means the long tail is on the left (mean < median).
Population	The entire group you want information about.
Sample	The part of the population you actually collect data from.
Parameter	A number that describes a population (for example μ , σ , or a population percentage).
Statistic	A number that describes a sample (for example $\bar{x}$, or a sample percentage).
Hypothesis	A claim about a characteristic of a population, which you test with data.
Random sample	Every member of the population has an equal chance of being chosen.
Unbiased sample	A sample that is representative of the population.
Biased sample	A sample that over-represents or under-represents part of the population.
Survey	Asking people questions to collect data.
Observational study	Watching or measuring individuals without changing anything.
Experiment	Imposing a treatment on individuals to see its effect. The only method that can show cause and effect.
Simulation	Using a model (coins, dice, random numbers, computer) to reproduce a real situation.
Control group	The group in an experiment that does not get the treatment (or gets a placebo), used for comparison.
Placebo	A harmless fake treatment, so subjects don't know which group they are in.

Memory trick: Population goes with **P**arameter; Sample goes with **S**tatistic.

Part 2 — Interpreting Normally Distributed Data

Key facts about a normal distribution:

- It is bell-shaped and symmetric about the mean μ .
- The mean, median and mode are equal, all at the center.
- The total area under the curve is 1, so the area over an interval is the **probability** (or the fraction of the data) in that interval.
- About 68% of the data is within 1 standard deviation of the mean, about 95% within 2, and about 99.7% within 3 (the **empirical rule**).



How to use the picture: mark μ in the middle, then add and subtract σ to label $\mu \pm \sigma$, $\mu \pm 2\sigma$ and $\mu \pm 3\sigma$. Then add up the percentages of the pieces you need.

Example 1 — Using the empirical rule

The heights of 10th-grade boys at a high school are normally distributed with mean 67 inches and standard deviation 2.5 inches.

(a) What percent of the boys are between 64.5 in and 69.5 in tall?

- $64.5 = 67 - 2.5 = \mu - \sigma$, and $69.5 = 67 + 2.5 = \mu + \sigma$.
- The interval $\mu - \sigma$ to $\mu + \sigma$ holds $34\% + 34\% = \mathbf{68\%}$.

(b) What is the probability that a randomly chosen boy is taller than 72 in?

- $72 = 67 + 2(2.5) = \mu + 2\sigma$.
- The part of the curve to the right of $\mu + 2\sigma$ is $2.35\% + 0.15\% = \mathbf{2.5\%}$, so the probability is 0.025.

(c) What percent are between 62 in and 69.5 in?

- $62 = 67 - 2(2.5) = \mu - 2\sigma$, and $69.5 = \mu + \sigma$.
- From $\mu - 2\sigma$ to μ is $13.5\% + 34\% = 47.5\%$. From μ to $\mu + \sigma$ is 34% .
- Total: $47.5\% + 34\% = \mathbf{81.5\%}$.

(d) Out of 300 boys, about how many are shorter than 64.5 in?

- $64.5 = \mu - \sigma$. The area to the left of $\mu - \sigma$ is $13.5\% + 2.35\% + 0.15\% = 16\%$.
- $0.16 \times 300 = \mathbf{48 \text{ boys}}$.

Part 3 — Using a z-Score and the Standard Normal Table

The empirical rule only works for values exactly 1, 2 or 3 standard deviations from the mean. For anything else, convert to a **z-score** and use the **standard normal table**.

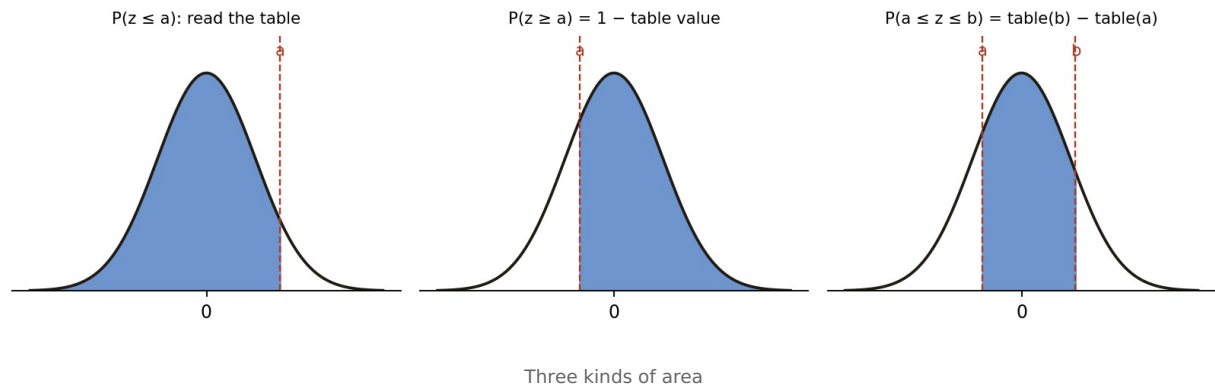
Why z-scores work. Subtracting μ moves the center to 0, and dividing by σ measures distance in “standard deviations”. This turns any normal distribution into the standard normal distribution (mean 0, standard deviation 1), so one table works for every problem.

Steps:

1. Compute $z = (x - \mu) / \sigma$. Round to two decimal places.
2. Look it up: the **row** is the ones and tenths digits, the **column** is the hundredths digit. The number in

the table is the area to the **left** of z .

3. Use the right rule for the question:



The question asks for...	Do this
"less than", "at most", "below"	Table value at z
"greater than", "at least", "above"	$1 - \text{table value at } z$
"between a and b "	Table value at $b - \text{table value at } a$

Example 2 — Finding probabilities with z-scores

SAT Math scores are normally distributed with mean 500 and standard deviation 100.

(a) What is the probability that a student scores 650 or less?

- $z = (650 - 500) / 100 = 150 / 100 = 1.50$.
- Table, row 1.5, column .00: **0.9332**.
- So about **93.3%** of students score 650 or less.
- Calculator check: `normalcdf(-1E99, 650, 500, 100)` = 0.9332.

(b) What is the probability that a student scores more than 430?

- $z = (430 - 500) / 100 = -70 / 100 = -0.70$.
- Table, row -0.7 , column .00: 0.2420. That is the area to the **left**.
- "More than" means the area to the **right**: $1 - 0.2420 = \mathbf{0.7580}$ (about 75.8%).

(c) What percent of students score between 420 and 610?

- z for 420: $(420 - 500) / 100 = -0.80$, and the table gives 0.2119.
- z for 610: $(610 - 500) / 100 = 1.10$, and the table gives 0.8643.
- Between: $0.8643 - 0.2119 = \mathbf{0.6524}$, so about **65.2%**.

Example 3 — Working backwards (a percentile)

Using the same SAT Math scores ($\mu = 500$, $\sigma = 100$): what score is at the **90th percentile**?

- We need the area to the left to be 0.90. Search **inside** the table for the value closest to 0.9000: it is 0.8997, in row 1.2, column .08, so $z \approx 1.28$.
- Convert back to a score: $x = \mu + z \cdot \sigma = 500 + 1.28 \times 100 = \mathbf{628}$.
- A score of about 628 beats 90% of test takers. Calculator check: `invNorm(0.90, 500, 100)` ≈ 628.2 .

Example 4 — Small-probability questions (negative z)

The battery life of a phone model is normally distributed with mean 42 months and standard deviation 6 months. What percent of batteries fail before 34.5 months?

- $z = (34.5 - 42) / 6 = -7.5 / 6 = -1.25$.

- Table, row -1.2 , column $.05$: **0.1056**.
- So about **10.6%** of batteries fail before 34.5 months.

Example 5 — Comparing scores with z-scores

Dhairya scored 82 on an English test (class mean 75, standard deviation 5) and 88 on a Math test (class mean 80, standard deviation 8). On which test did he do better **compared with his class**?

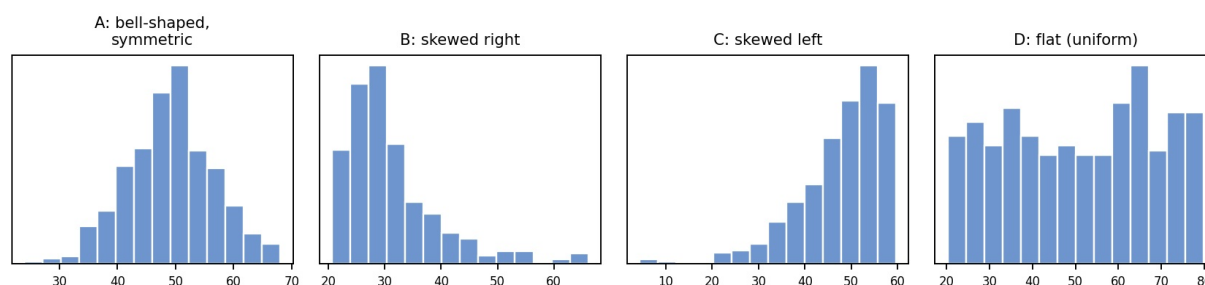
- English: $z = (82 - 75) / 5 = 1.40$.
- Math: $z = (88 - 80) / 8 = 1.00$.
- The English score is 1.4 standard deviations above the mean, and the Math score only 1.0. So **English** was the relatively better performance, even though the Math score is the higher number.

Part 4 — Recognizing Normal Distributions

Checklist: data is approximately normal when its histogram is:

1. **Single-peaked** (one hump) in the middle,
2. **Symmetric**: the left half roughly mirrors the right half, and
3. **Bell-shaped**: it tapers off evenly on both sides, with most data near the center and very little far out.

The quick number test: for normal data, **mean $\approx$ median**. If the mean is much bigger than the median, the data is skewed right; if it is much smaller, the data is skewed left.



Which histogram is normal?

Histogram	Shape	Normal?	Why
A	Bell-shaped, symmetric	Yes	One central peak, even tails on both sides
B	Skewed right	No	Long tail to the right (e.g. household incomes)
C	Skewed left	No	Long tail to the left (e.g. scores on an easy test)
D	Uniform (flat)	No	No peak; every value about equally common

Example 6 — Is it normal?

(a) The number of minutes students spend on social media per day has a mean of 128 minutes and a median of 95 minutes. Is this likely a normal distribution?

- In a normal distribution the mean equals the median. Here the mean is much larger than the median.
- A few students with very high screen time pull the mean up, which means the data is **skewed right**. So it is **not normal**, and the empirical rule and the z-table should not be used on it.

(b) The weights of 500 bags of chips have a histogram that is bell-shaped and symmetric, with mean 9.02

oz and median 9.01 oz. Is it reasonable to use the standard normal table on these weights?

- **Yes.** The shape is bell-shaped and symmetric, and the mean is almost equal to the median, so the data is approximately normal.

Part 5 — Populations vs. Samples, Parameters vs. Statistics

	Population	Sample
What it is	The whole group you want to know about	The part of the group you actually study
A number describing it is a	Parameter	Statistic
Symbols	μ (mean), σ (standard deviation), p (proportion)	$\bar{x}$ (mean), s (standard deviation), p (proportion)
Usually known?	Usually not : too big to measure all of it	Yes : we measure it
Changes from sample to sample?	No: it is one fixed number	Yes: a different sample gives a different value

How to decide in a word problem:

- Ask “who does the question want to know about?” That group is the **population**.
- Ask “who was actually measured or asked?” That group is the **sample**.
- If the number comes from **everyone** in the population, it is a **parameter**. If it comes from only **some** of them, it is a **statistic**.

Example 7 — Identify the population and the sample

(a) A survey of 1,200 randomly selected U.S. adults found that 64% drink coffee daily.

- **Population:** all adults in the United States.
- **Sample:** the 1,200 adults surveyed.

(b) To check the quality of a day’s production, a factory tests 50 of the 4,000 light bulbs it made that day.

- **Population:** all 4,000 bulbs made that day.
- **Sample:** the 50 bulbs tested.

Example 8 — Parameter or statistic?

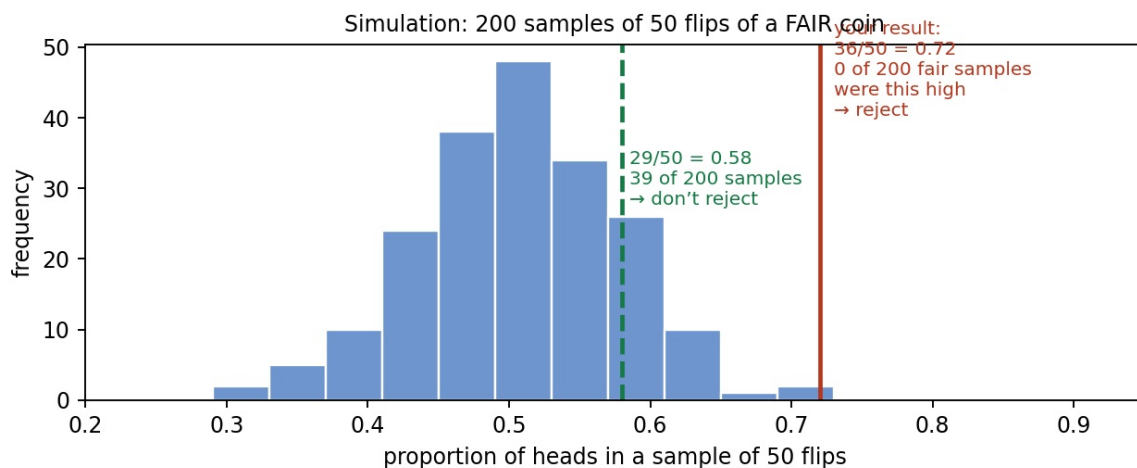
Statement	Answer	Reason
The mean age of all 32 students in Dhairya’s class is 14.6 years.	Parameter	It describes the whole group being studied (the class).
In a poll of 800 voters, 52% support the new park.	Statistic	It describes only the 800 voters sampled, not all voters.
The 2020 Census counted all U.S. residents and found 16.8% were 65 or older.	Parameter	A census measures the entire population.
A random sample of 40 NBA players has a mean height of 6 ft 6.5 in.	Statistic	It comes from 40 players, not every player.

Part 6 — Analyzing a Hypothesis

A **hypothesis** is a claim about a population. We can’t check the whole population, so we compare our **sample result** with what would usually happen **if the claim were true**.

Steps:

1. **State the hypothesis** (the claim). Example: “This coin is fair”, i.e. heads comes up 50% of the time.
2. **Collect a sample** and find its statistic. Example: 50 flips, 36 heads, so $36/50 = 0.72$.
3. **Simulate** many samples of the same size, *assuming the claim is true*. Example: 200 samples of 50 flips of a fair coin.
4. **Compare**. How often did the simulation produce a result as extreme as yours?
 - **Rarely** (roughly 5% of the time or less): your result would be very unlikely if the claim were true, so **reject the hypothesis**.
 - **Often**: your result is normal for a true claim, so **don't reject** it. This does not prove the claim; it only means the data doesn't contradict it.



Simulation of a fair coin

Example 9 — A suspicious coin (reject)

You flip a coin 50 times and get 36 heads. You claim the coin is fair. Should you reject the claim?

- Sample proportion: $36 / 50 = 0.72$.
- In the simulation of 200 samples of 50 fair flips, **none** reached 0.72. The simulated results cluster around 0.50, and almost all fall between 0.36 and 0.64.
- Getting 0.72 from a fair coin would be extremely unlikely (the exact probability is about 0.1%).
- **Conclusion: reject the hypothesis.** The coin is probably not fair.

Example 10 — An ordinary result (don't reject)

A friend flips the same kind of coin 50 times and gets 29 heads (0.58). Reject the claim that it is fair?

- In the simulation, 39 of the 200 samples (19.5%) were 0.58 or higher.
- That is common, far more than 5%, so this result is quite possible with a fair coin.
- **Conclusion: do not reject the hypothesis.** There isn't enough evidence that the coin is unfair.

Part 7 — Methods of Data Collection, Sampling & Bias

7A. The four methods of collecting data

Method	What the researcher does	Changes anything?	Can show cause and effect?	Example	Clue words
Survey	Asks people questions	No	No	Asking 200 students how many hours they	“asked”, “poll”, “questionnaire”, “responded”

				sleep	
Observational study	Watches or measures individuals as they are	No	No, only an association	Recording how many drivers wear seat belts at an intersection	"observed", "recorded", "tracked", "compared people who already..."
Experiment	Imposes a treatment on one group and compares with a control group	Yes	Yes , if the groups are randomly assigned	Randomly giving half the tomato plants a new fertilizer and comparing growth	"randomly assigned", "treatment", "control group", "placebo"
Simulation	Uses a model to imitate a real situation	Nothing real	No: it models, it doesn't measure	Using random numbers to model 1,000 free throws by a 70% shooter	"model", "random number generator", "simulate", "coin/die/spinner"

The key difference between an observational study and an experiment: in an observational study the individuals **choose** their own group (people who already exercise vs. people who don't). In an experiment the **researcher assigns** the groups. That is why only an experiment can prove cause and effect.

A good experiment has:

- a **control group** (no treatment or a placebo),
- **random assignment** of subjects to groups, and
- enough subjects (**replication**), so the result isn't down to luck.

Example 11 — Identify the method

Situation	Method	Reason
A researcher records how long each shopper waits in line at a grocery store.	Observational study	Measures what happens without changing anything
The school asks every 10th student entering the cafeteria to rate the lunch menu.	Survey	Asks people questions
60 volunteers are randomly split: 30 take a vitamin, 30 take a placebo. Their colds are counted for 3 months.	Experiment	A treatment is imposed, with a control group and random assignment
A student uses a random number generator to model rolling two dice 500 times.	Simulation	A model imitates a real process
A study compares test scores of students who chose to take music lessons with those who didn't.	Observational study	The students picked their own group; nothing was imposed

7B. Sampling methods (how the sample is chosen)

Method	How it is done	Example	Biased?
Random	Every member has an equal chance (names from a hat, random number generator)	Randomly pick 100 student ID numbers	Unbiased (best)
Stratified	Split the population into groups (strata), then randomly pick from each group	Randomly choose 25 students from each of grades 9, 10, 11 and 12	Unbiased
	Use a rule, like every 10th	Survey every 10th person	

Systematic	Use a rule, like every 10th member of a list	Survey every 10th person entering the stadium	Usually unbiased
Cluster	Split into groups, randomly pick whole groups, and use everyone in them	Randomly choose 3 homerooms and survey every student in them	Usually unbiased
Convenience	Pick whoever is easiest to reach	Survey the students sitting at your lunch table	Biased
Self-selected	Members volunteer to respond	An online poll anyone can answer	Biased : people with strong opinions respond most

Stratified vs. cluster, the most common mix-up:

- **Stratified**: a **few from every** group.
- **Cluster**: **all from a few** groups.

7C. Identifying bias

A sample (or a question) is **biased** if it systematically favors certain results.

Type of bias	What goes wrong	Example
Sampling bias (unrepresentative sample)	The sample leaves out, or over-represents, part of the population	Surveying people leaving a gym to estimate how often U.S. adults exercise
Self-selection / voluntary response	Only people who care strongly answer	A TV station's call-in poll about a new tax
Nonresponse bias	Many people chosen don't reply, and they may differ from those who do	Mailing 1,000 surveys and getting only 90 back
Question (wording) bias	The question pushes toward an answer	"Don't you agree that the unfair new curfew should be removed?"

Example 12 — Find the bias, then fix it

(a) To estimate how much time U.S. teens spend on video games, a researcher surveys 150 teens at a gaming convention.

- **Biased: sampling bias.** Teens at a gaming convention likely play far more than typical teens.
- **Fix:** take a random sample of teens from many schools, for example a stratified random sample from each region of the U.S.

(b) A principal wants students' opinions on the dress code and posts an optional online form.

- **Biased: self-selected.** Mostly students who strongly dislike (or strongly like) the dress code will respond.
- **Fix:** randomly select students from the full school roster and ask each of them.

(c) Survey question: "Should the city waste money on a new skate park?"

- **Biased: question wording.** The word "waste" pushes people toward "no".
- **Fix:** "Do you support or oppose building a new skate park?"

(d) A teacher surveys the 12 students in the front two rows about whether homework should be reduced.

- **Biased: convenience sample.** Students who sit in front may not represent the class.
- **Fix:** use a random number generator to choose students from the whole class roster.

Part 8 — Evaluating Published Reports

When you read a claim in a news story or advertisement, check these five things:

1. **Who was studied?** Is the sample **random** and **large enough**, and does it represent the population the claim is about?
2. **How was the data collected?** Survey, observational study or experiment? **Only an experiment** supports a cause-and-effect claim.
3. **Is there bias?** Look for convenience or self-selected samples, low response, loaded questions, or a sponsor who benefits from the result.
4. **What is the margin of error?** Use $\pm 1/\sqrt{n}$. If the difference being claimed is smaller than the margin of error, the claim is not supported.
5. **Does the conclusion match the data?** Watch for claims about “all Americans” from a sample of one school, or “causes” from a study that only observed.

Example 13 — An advertisement

“4 out of 5 dentists recommend Brite-Smile toothpaste!” The fine print: the company sent a survey to 25 dentists who already stock its product; 15 replied and 12 of them recommended it.

- **Sample:** very small (15 replies), and only dentists who **already sell** the product, so it is a biased sample.
- **Nonresponse:** 10 of the 25 dentists didn’t reply.
- **Sponsor:** the company benefits from a positive result.
- **Conclusion:** the claim is **not reliable**. “4 out of 5” (12 of 15) does not represent dentists in general.

Example 14 — “Causes” from an observational study

“Study shows eating breakfast raises test scores!” Researchers recorded whether 2,000 students ate breakfast and compared their grades. Breakfast eaters averaged higher.

- **Method:** observational study. Nobody was **assigned** to eat breakfast; students chose.
- Other factors could explain the difference: sleep, family routines, study habits.
- **Conclusion:** the data shows an **association**, not cause and effect. To show cause, you would need an experiment that randomly assigns students to eat or skip breakfast.

Part 9 — Estimating a Population Mean (and Proportion)

We usually can’t measure a whole population, so we use a **sample statistic to estimate the population parameter**:

- use the **sample mean $\bar{x}$** to estimate the **population mean μ** ;
- use the **sample proportion $\hat{p}$** to estimate the **population proportion p** .

Why it works, and how to make it better:

- A **random** sample tends to look like the population, so its mean lands close to μ .
- Different samples give slightly different means. This is called **sampling variability**.
- **Larger samples vary less**, so their means are closer to μ . More data gives a better estimate.
- Taking **several** random samples and averaging their means gives an even better estimate.

Example 15 — Estimate a population mean from one sample

A random sample of 10 students at a high school recorded how many minutes they spent on homework last night:

45, 60, 30, 50, 55, 40, 65, 35, 50, 70

Estimate the mean homework time for **all** students at the school.

- Add: $45 + 60 + 30 + 50 + 55 + 40 + 65 + 35 + 50 + 70 = 500$.
- Divide: $\bar{x} = 500 / 10 = \mathbf{50 \text{ minutes}}$.

- The sample mean is a **statistic**. It is our estimate of the population mean μ (a **parameter**), so the best estimate is that students spend about **50 minutes**.

Example 16 — Several samples

Four more random samples of 10 students each gave means of 53, 48, 47 and 52 minutes. Together with Example 15's 50 minutes, what is a better estimate of μ , and what does the spread tell you?

- Mean of the five sample means: $(50 + 53 + 48 + 47 + 52) / 5 = 250 / 5 = \mathbf{50 \text{ minutes}}$.
- The sample means range from 47 to 53, so a single sample of 10 can be off by about 3 minutes.
- Using samples of **40** students instead would make the means cluster **more tightly** around the true value.

Example 17 — Margin of error in a poll

In a random sample of 900 likely voters, 52% said they will vote for Candidate A. Can the newspaper say Candidate A will win a majority?

- Margin of error: $\pm 1/\sqrt{900} = \pm 1/30 \approx \pm 0.033$, which is **$\pm 3.3\%$** .
- Interval: $52\% - 3.3\% = 48.7\%$ up to $52\% + 3.3\% = 55.3\%$.
- The interval includes values **below 50%**, so the true support could be less than half.
- **Conclusion: no**, the poll cannot say Candidate A will win a majority. The race is too close to call.

Example 18 — How big a sample?

A school wants a margin of error of no more than $\pm 5\%$. How many students should it survey?

- Set $1/\sqrt{n} = 0.05$, so $\sqrt{n} = 1 / 0.05 = 20$, and $n = 20^2 = \mathbf{400 \text{ students}}$.

Part 10 — Last-Minute Check (answers below)

1. A normal distribution has $\mu = 80$ and $\sigma = 6$. What percent of the data lies between 68 and 92?
2. For the same distribution, find z for $x = 71$.
3. Using the table, what is $P(z \leq -1.50)$?
4. "A survey of 500 of the 3,000 employees found that 38% bike to work." Is 38% a parameter or a statistic?
5. Random sampling from each grade level is which sampling method?
6. Researchers **randomly assign** 50 people to a new diet and 50 to their usual diet. Survey, observational study or experiment?
7. "Wouldn't you agree that our amazing mayor deserves re-election?" What kind of bias is this?
8. A random sample of 625 people is polled. What is the margin of error?
9. The mean of a data set is 30 and the median is 44. Is it skewed left, skewed right, or normal?
10. You roll a die 60 times and get 40 sixes. Should you reject the claim that the die is fair? Why?

Answers:

1. $68 = \mu - 2\sigma$ and $92 = \mu + 2\sigma$, so **95%**.
2. $z = (71 - 80) / 6 = \mathbf{-1.5}$.
3. Table, row -1.5 , column $.00$: **0.0668**.
4. **Statistic**. It describes the 500 employees sampled, not all 3,000.
5. **Stratified** sampling.
6. **Experiment**. A treatment is imposed and the groups are randomly assigned.
7. **Question (wording) bias**. "Amazing" pushes toward yes.
8. $1/\sqrt{625} = 1/25 = \mathbf{\pm 4\%}$.
9. The mean is less than the median, so it is **skewed left**.
10. **Reject**. A fair die gives a six about $1/6$ of the time, about 10 in 60 rolls. Forty sixes would almost

never happen with a fair die.

Standard Normal Table (area to the LEFT of z)

How to read it: for $z = -1.25$, go to row **-1.2** and column **.05**, which gives 0.1056. For $z = 1.28$, go to row **1.2** and column **.08**, which gives 0.8997.

Negative z: area to the LEFT of z (e.g. $z = -1.25 \rightarrow$ row **-1.2**, column **.05**)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Standard normal table, negative z

Positive z: area to the LEFT of z (e.g. z = 1.28 → row 1.2, column .08)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Standard normal table, positive z